Sandesh Lamichhane

sandesh.lamichhane1@gmail.com · linkedin.com/in/sandesh-l · sandesh-l.com

Experience

Systems Professional II / Senior Developer — University of Kentucky (ITS RCI)
Lexington, KY | July 2024 – Present

- Designed and shipped an HPC Account Management Portal end-to-end (**Python/Django** backend with **MySQL**, **Angular** frontend, **OIDC/CILogon SSO**). Implemented a split-server **OAuth authorization-code flow**: frontend handles user auth with **nonce verification**, backend independently validates the **JWT** and user-DB state before issuing scoped session data. Debugged production token-flow issues including expiry, missing claims, and nonce mismatch. Covers allocations, usage, and project-membership flows for users, PIs, admins, directors, and staff.
- Designed and built the ACCESS Software Documentation Service as a multi-tier Python data platform (**Peewee** ORM throughout): an institution-agnostic **FastAPI** core API, an institution-aware ACCESS-SDS tier, and **Flask/Gunicorn** consumer web layer. Powers **4+ first-party ACCESS apps** and multiple third-party integrations; catalogs **3,974 packages**.
- Designed and built an **LLM**-driven self-healing data-growth pipeline for SDS: on-demand metadata backfill for unknown software, alias flagging that routes consumer-reported corrections upstream, and human-in-the-loop review gating all AI-generated content before publication.
- Automated HPC cluster builds via **Ansible**, **OpenHPC**, and **Warewulf** provisioning on **Rocky Linux** images, plus containerized core infrastructure services. Cut cluster build time by **25%** and maintenance/downtime windows by **50%**.
- **CI/CD** for the HPC Account Portal via **GitHub Actions**: automated test suite gates pushes to a dev server for manual validation, with controlled promotion to production.
- Co-operate three production HPC clusters (~**32,000 cores**, ~**5 PB GPFS**) serving **1,200+ active users across 55+ departments**, with a small team, including on-call rotation. Responsibilities span hardware lifecycle, Slurm/GPFS upgrades, security patching, and capacity planning.
- Built and operate a **Prometheus + Grafana** telemetry pipeline across cluster nodes and network fabric. Alert signals drove isolation of overheat-prone hardware into clearly labeled queues, identification and resolution of network topology mismatches, and informed capacity / upgrade-purchase decisions.
- Operate under role-scoped privileged-access policies: **key-only authentication**, team-scoped **sudo** (e.g., user-facing team scoped to login nodes; networking team scoped to a defined command set cluster-wide), full audit logging of privileged actions. User auth federated via **LDAP** plus **SSH keys**, with public keys published to a central management DB gated behind **CILogon OAuth + 2FA**; key edits notify operators and central removal cascades to cluster key teardown. (Operate within these policies; not the original implementer for most.)
- Contributed to live-site response on multiple high-severity Linux/HPC CVEs including **Copy-Fail** (CVE-2026-31431, disclosed 2026-04-29) and **Dirty-Frag**: monitored cluster posture through patch rollout and validated cluster recovery.
- Operate two NVIDIA GPU clusters spanning **V100**, **P100**, **H100**, and **H200** generations; assist with NVIDIA driver installs and upgrades; troubleshoot GPU and host-side issues, including debugging **CUDA-aware Python** libraries in MPI contexts.
- Configure and support the **InfiniBand** fabric (**IPoIB**, **RDMA**): traffic monitoring, leaf-switch topology tuning to minimize IB hops for **MPI** workloads, and ongoing troubleshooting.
- Configure **MIG** partitioning of NVIDIA GPUs at cluster build/reconfiguration, exposing partitions as separate **Slurm** nodes to improve cluster utilization.
- Operate the platform's container runtimes: **Singularity/Apptainer** for user containers, **Podman** for system containers, **Docker** for off-cluster applications.
- Co-architected one HPC cluster (**Slurm**, **GPFS**, isolated networking), including rack-level power budgeting and redundant PDU planning (limited scope); contributed to the architecture of a second.
- Built a cross-cluster instrument access pipeline supporting **20 connected instruments**, with low-overhead onboarding for new ones. Integrates LibreBooking scheduling, Slurm orchestration, Windows-host execution (VNC + networking), and automated post-session data movement to GPFS. User entry point is a custom **Open OnDemand** portal app: users join live sessions with one click in-browser, no CLI required.
- Designed and built additional custom **Open OnDemand** portal apps (browser-based job and session entry points). Wired OOD authentication through **CILogon** on current systems and **LDAP** for older systems.
- Lead a **10-person student dev team** (graduate + undergraduate); mentor newer staff and serve on student-developer interview and hiring panels.
- Team contributions span the ACCESS Resource Advisor, Software Documentation Service, and current HPC data pipelines for RAG/MCP tooling. **2 peer-reviewed conference papers** from this work (DOIs `10.1145/3708035.3736100`, `10.1145/3708035.3736098`).
- Contributing to **LLM/RAG and MCP-tool** infrastructure for the production ACCESS chatbot ecosystem.

- Primary lead on a **Model Context Protocol (MCP)** integration for the Software Documentation Service, exposing the 3,974-package catalog as a tool surface for LLM agents; redesigned return schemas, added query parameters, and built **fuzzy API search** to fit LLM consumer needs.

ACCESS Track 2 Software Developer — University of Kentucky (NSF)
Lexington, KY | September 2022 – June 2024

- Designed and built the ACCESS Resource Advisor recommender system from scratch. Helps prospective users identify which of ACCESS's **15+** national HPC resources best fits their workload; **500+ recommendations** generated to date. Later iterated on the frontend UX based on user feedback.
- Designed and built the initial ACCESS Software Documentation Service (SDS), surfacing **1,500+ software packages** across HPC resources via **REST APIs**.
- Designed the software architecture and SQL data layer underpinning the recommender backend.
- Built REST API integrations (including **Confluence**) to keep HPC-system metadata current and queryable from end-user tools.
- Authored cross-system login and onboarding documentation across multiple HPC resources.

Software Developer — Berea College
Berea, KY | July 2021 – May 2022

- Built full-stack applications for multiple Berea College departments: volunteer/service-learning hour tracking, course-catalog management, and chemical inventory. Stack: **Python**, **Flask**, **MySQL**, **Bootstrap**; **MVC** architecture with unit-test coverage.
- Improved accessibility (screen-reader support, color/text/icon contrast) and cross-device compatibility across the application portfolio.

Open Source Contributor — Runestone Interactive
Remote | June 2020 – August 2020

- Co-authored the first edition of *How to Think Like a Data Scientist* (open-source textbook on Runestone Academy). Wrote business datasets, data-cleaning lessons, and instructional content covering **pandas**, **scikit-learn**, and introductory ML / predictive modeling. (A later edition retains portions of this work; not involved in that revision.)
- Reworked book content for **Google Colab** to remove install friction for remote learners.

Projects

LOOM — Cross-Domain Trend Detection System (in progress)
Python 3.12, async SQLAlchemy 2.0 + Postgres (asyncpg), Pydantic v2, Pyright strict, ruff, uv | 2026-01 – Present

- Designed a **bitemporal data model** from scratch: explicit valid-time / transaction-time separation, supersedes-chain for revisable state, as-of replay queries enabling historical reconstruction and prediction-calibration scoring. ~**360 integration tests against real Postgres** (no DB mocking, by design).
- Built a statistical drift-detection pipeline over periodic graph snapshots: **Jensen-Shannon divergence** on entity/topic distributions, **CUSUM** change-point detection, cross-domain bridge analysis. Detection stays statistics-first; LLM use deferred to a separate interpretation layer.
- Architected **reversibility-first semantics**: provenance (origin, method, confidence) on every node, version tables capturing pre-state on every mutating write, **Beta priors with Wilson 95% lower-bound** for source-trust defaults.

ML Inference & Edge Models — Personal Projects
PyTorch, TensorFlow/tflite, Keras, ONNX, Edge-device deployment

- Built and adapted inference pipelines across speech, vision, and gesture using pretrained foundation models: **Whisper tiny, sherpa, and wav2vec** (speech-to-text), **MediaPipe** (gesture recognition and hand-landmark detection), and **Meta SAM2 and SAM3D** (image segmentation).
- Cross-toolchain model conversion via **ONNX** between **PyTorch** and **TensorFlow/tflite** to match existing target runtimes.
- Applied **fine-tuning** and **model pruning** to adapt pretrained models and reduce inference cost for edge-device deployment.

MyPariwar — Full-stack genealogy app
Go, SQLite, React, TypeScript, Tailwind CSS, Cloudflare R2

- Built a full-stack genealogy app with a **pure-Go backend (no CGo)**, SQLite persistence via a sqlc-generated query layer, a React/TypeScript frontend, and auth-gated photo storage on **Cloudflare R2**.
- Designed an audit logging system using **SQLite** triggers (deletes) and middleware (inserts/updates). Tracks complete change history without modifying query logic.
- **Auth**: short-lived **JWT** access tokens (HS256, 15 min) plus rotated refresh tokens (32-byte random, **SHA-256-hashed at rest**, delivered as **HttpOnly / Secure / SameSite=Strict** cookies scoped to `/api/auth`, 30-day expiry). Concurrent-401 refresh requests coalesced client-side. Session revocation on logout, password change, and admin deactivation; refresh endpoint enforces `is_active`. Revoked tokens retained with `revoked_at` for audit trail.
- **Hardening**: **bcrypt** password storage (cost 10), invite-only registration with 7-day hashed tokens (atomic create-user + mark-used transaction), generic "invalid credentials" error to avoid email/password disclosure, sliding-window rate limit keyed on `CF-Connecting-IP` (10 req/min on login/register/password-change). JWT secret length-validated (32+ bytes, not a known default); expired refresh tokens cleaned up at startup.
- Deployed on a **Hetzner VPS** behind **Cloudflare Tunnel** with **Tailscale**. Zero exposed ports, automated SQLite backups.
- **CI/CD** via **GitHub Actions**: automated tests on every push, gating promotion to a dev server for manual validation and then controlled push to production.

Improved Models for Health-Related Social Media Text Classification

NLP / ML | github.com/Sandesh-L/NLP-project

- Increased F1 score by **31 points (64 → 95)** over prior published results on the NPMU health-text dataset.
- Custom data-cleaning pipeline, learning-rate tuning, and pretrained-model selection via comparative experiments.

ABC to TAB — Automated Guitar Tablature Generation

AI planning / Python | github.com/Sandesh-L/abc2tab

- Built a system to convert ABC music notation into guitar tablature with optimized note placement.
- Integrated abc2abc parser, a custom guitar model, and **Fast Downward planning (LM-Cut heuristic)**.
- Conducted user testing to validate playability and tablature quality.

Selected Freelance

- Opal Events & Arrangements: responsive client site for an event-planning business; services showcase with clean-design focus. **Vue**, **Tailwind**, **Vercel**.
- Garage Door Solutions of KY: SEO-focused business site with services and service-area pages, responsive UI. **Astro**, **JavaScript**.

Education

M.S., Computer Science — University of Kentucky
Lexington, KY | Aug. 2022 – May 2024

B.A., Computer & Information Sciences and Mathematics (double major) — Berea College
Berea, KY | Aug. 2018 – May 2022

Skills

Languages: Python, Go, C, C++, SQL, Bash
Databases: PostgreSQL, MySQL, SQLite
Web & Frameworks: JavaScript, TypeScript, React, Angular, Vue, Node.js, FastAPI, Flask, Gunicorn, Django, GraphQL, Tailwind CSS
Python Tooling: uv, pyproject.toml, Pydantic v2, Pyright, ruff, structlog, conda/miniforge, SQLAlchemy 2.0 (async), Peewee, Alembic
Auth & Security: OIDC, OAuth 2.0, CILogon, Keycloak, JWT, bcrypt, 2FA federation
HPC & Infrastructure: Slurm, OpenHPC, Warewulf, GPFS, Ansible, Singularity/Apptainer, Podman, Docker, Kubernetes, MPI, InfiniBand (IPoIB, RDMA), Open OnDemand, Lmod, Grafana, Prometheus
GPU & ML: NVIDIA GPUs (V100/P100/H100/H200), NVIDIA drivers, MIG, PyTorch, TensorFlow/tflite, Keras, ONNX, model fine-tuning & pruning, vLLM, NLP
Tools & Platforms: Linux (Rocky / RHEL), Git/GitHub, AWS, Azure (AI, Copilot, HPC offerings; exposure via UKy–Microsoft partnership), Cloudflare, Tailscale, Netbox, Jira, Confluence